

Field Consistency: The Cost of Enforcing PDE Residuals on 2D Fields

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TABLE I

FIELD-LEVEL GENERALIST VS. PER-INSTANCE PINN (DEEPIXDE).
A SPECIALIST WINS ON ITS OWN PROBLEM; THE GENERALIST IS A
CONSISTENT SOLVER OF ALL INSTANCES.

Instance	DeepXDE	generalist
instances	–	–

III. RESULTS

REFERENCES

- [1] L. Lu, X. Meng, Z. Mao, and G. E. Karniadakis, “DeepXDE: a deep learning library for solving differential equations,” SIAM Review, vol. 63, pp. 208–228, 2021.

Abstract—The tensor pipeline extends to 2D fields: the same transformer with a field-shaped head learns the full $u(x, y)$ surface of a heat plate. The showcase number – 5.9% peak field error – is real and is the easiest member of the validation distribution, whose honest mean is 29%; this paper reports the distribution, not just the showcase. Enforcing the Poisson residual cuts it 6–8 $\times$ while field fidelity degrades: consistency is not accuracy at the field level. Against a per-instance PINN (DeepXDE, 19 minutes per problem), the specialist wins on its own instance at 0.03–0.06% error but with a higher governing-equation residual; the generalist answers any instance in one forward pass. Both statements are true; the claim is the tradeoff.

Index Terms—physics-informed neural networks; PDEs; PINN; fields; verification

I. INTRODUCTION

At the field level the pattern from trajectory domains repeats and sharpens: enforcing the governing equation makes predictions consistent, not correct. The two must be measured separately, and here both are – including the honest distribution number that a showcase-only evaluation hides.

II. PROTOCOL

Heat-plate domain ($u_{xx} + u_{yy} = f$), field-shaped head, 240 epochs, canonical showcase plus the full validation distribution; Burgers’ equation for the per-instance DeepXDE comparison (per-instance PINN at 0.03–0.06% full-field error vs. one-pass generalist at 33–51% with lower governing-equation residual).